

FPGA Acceleration for Urban Digital Twin Models: Optimizing Mixed-Integer Linear Programming (MILP) for Energy Efficiency

Project Objectives

The motivation of this project is to design and develop novel energy-efficient hardware accelerators for solving MILP problems on FPGA platforms. For fast prototyping, we will adopt a hardware-software co-design approach to explore novel domain-specific accelerators using High Level Synthesis. By leveraging the parallelism and reconfigurability of FPGAs, the goal is to achieve significant energy savings while maintaining or improving the computational performance of MILP solvers.

Obligatory Objectives

1. **Understand and Describe MILP Kernels:**
 - Conduct an in-depth analysis of key MILP algorithms, focusing on the most computationally intensive kernels.
 - Identify the parts of the MILP solving process that can benefit most from hardware acceleration.
2. **Implement and Optimize MILP Kernels on FPGA Using High-Level Synthesis (HLS):**
 - Design and implement at least two critical kernels of the MILP solving process using HLS tools.
 - Focus on optimizing these kernels for energy efficiency, resource usage, and latency.
3. **Evaluate and Explore Design Trade-offs:**
 - Conduct a comprehensive design space exploration to evaluate the trade-offs between energy consumption, computational latency, resource usage, throughput and hypermeters of the solver.
 - Present at least three Pareto optimal solutions for each implemented kernel, demonstrating the balance between these metrics.
4. **Experimental Evaluation on FPGA:**
 - Test and validate the designed accelerators on a selected FPGA platform.
 - Measure and compare energy consumption, execution time, and solution accuracy against traditional CPU-based implementations.
 - [Optional] Experimental validation of the proposed designs on UrbanTwin MILP problems, showcasing the advantages of FPGA-based acceleration.

Additional Objectives

1. **Integration with Full MILP Solver Pipeline:**
 - Integrate the optimized MILP kernels into a complete MILP solver pipeline on FPGA.
 - Evaluate the performance of the integrated system in terms of energy efficiency and computational throughput.
2. **Qualitative Analysis of Solution Accuracy:**
 - Assess the accuracy and quality of the solutions provided by the FPGA-accelerated MILP solver.
 - Compare the results with those obtained from traditional software-based solvers, focusing on metrics such as solution optimality and robustness.

Required Knowledge and Skills

- Proven experience in hardware description languages such as VHDL or Verilog.
- Experience with Xilinx FPGAs or Intel FPGAs and associated development tools.
- Strong programming skills in Python, C/C++, and familiarity with High-Level Synthesis (HLS) tools.
- Understanding of optimization techniques, particularly MILP.

Type of Work

- **Theoretical Analysis (30%):** Involves understanding MILP algorithms, energy-efficient design principles, and FPGA architecture.
- **Design and Experimentation (70%):** Focuses on the implementation, optimization, and evaluation of the MILP kernels on FPGA.

This project will be conducted under the supervision of experts in hardware design and optimization from the Embedded Systems Laboratory (ESL) and Industrial Process and Energy Systems Engineering (IPESE), providing the student with an opportunity to contribute to cutting-edge research in energy-efficient computing.

References

- Renewable Energy Hub Optimizer (REHO): <https://reho.readthedocs.io/en/main/>
- MILP with Gurobi (commercial): <https://www.gurobi.com/resources/chapter-1-why-mixed-integer-programming-mip/>
- HiGHS (open source): <https://highs.dev/>
- Vitis HLS User Guide: <https://docs.amd.com/r/en-US/ug1399-vitis-hls>

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